

## Questions

### Encapsulation

- 1. Create a BankAccount class with a private balance attribute.**
  - Add methods to deposit, withdraw, and display the balance.
- 2. Create a Student class with private attributes for name and marks.**
  - Use getter and setter methods to access and modify the marks.

### Abstraction

- 3. Create an abstract class Shape with an abstract method area().**
  - Implement subclasses Circle and Rectangle.
- 4. Create an abstract class Vehicle with an abstract method start\_engine().**
  - Implement the method in Car and Bike classes.

### Inheritance

- 5. Create a base class Animal with a method speak().**
  - Derive Dog and Cat classes and add their own methods.
- 6. Create a class Employee with attributes name and salary.**
  - Create a subclass Manager that adds a department attribute.
- 7. Demonstrate multilevel inheritance using classes Person → Employee → Developer.**
  - Add appropriate attributes and methods at each level.

### Polymorphism

- 8. Create classes Dog, Cat, and Cow, each having a sound() method.**
  - Write a function that calls sound() on any animal object.
- 9. Create classes CreditCard and UPI with a common method pay(amount).**
  - Show how the same method behaves differently for each payment type.

## Combined OOP Concepts

### 10. Design a simple Library Management System:

- Use:
  - **Encapsulation** for book details.
  - **Abstraction** for a generic library item.
  - **Inheritance** for Book and Magazine.
  - **Polymorphism** for displaying item information.